Inherited F4 eager rewrite through shift runs — run report

Run `run-20261016T2335Z` · binary `f6_ic_probe` built from `1750ec759` (local branch on `f25d5e520`, with the gate-first default) · 2026-10-16

What ran

`scripts/run_arms.sh`, one binary, arms by environment only: **bmi2** (default on this host: eager rewrite, `pdep ∘ pext` per segment), **runs** (`KIC_F4_EAGER=runs`: eager rewrite through shift runs, what a host without BMI2 now runs), **lazy** (`KIC_F4_EAGER=0`: lazy word-wise rewrite), **scatter** (`KIC_F4_REWRITE=scatter`: lazy rewrite with the bit scatter, what a host without BMI2 ran before this change). Nine cells as in `research/f6_gate_first_x86_20261016`, `--engines f4,f6p --reps 3`, 5 rounds, arm order rotated. Host: Intel Xeon @ 2.10 GHz, 4 vCPU KVM, BMI2, shared, unpinned, L0, load 1.01–1.92. 180 arm runs, all exited 0. No aarch64 host was available: **scatter** and **runs** are the before and after of a non-BMI2 host, measured here on x86.

Stage unit = per-arm total `word_xors`; CPU = probe `cpu_ms` (three repetitions). Ratios derived as in the gate-first report (medians over rounds; bracket = worst–best round pairing).

Exactness

All 80 targets have the same decision tuple in every arm and round. Stage units are identical across rounds; **runs** charges exactly **bmi2**'s units and **lazy** exactly **scatter**'s, on every cell and engine.

Measured values

cell	engine	units runs = bmi2	units scatter = lazy	CPU scatter ms	CPU runs ms	CPU bmi2 ms	scatter/runs	bmi2/runs
K1_17	f4	18019804	15109049	588.7	574.0	449.9	1.03 (0.74–1.23)	0.78
K1_17	f6p	10830979	8865230	193.9	183.0	148.9	1.06 (0.78–1.68)	0.81
K1_23	f4	62213299	52572523	1279.1	920.5	805.9	1.39 (1.18–1.76)	0.88
K1_23	f6p	38122485	32069197	454.4	340.8	333.1	1.33 (0.96–1.70)	0.98
K0_23	f4	167017420	148425559	3745.3	2723.4	2447.7	1.38 (1.03–1.57)	0.90
K0_23	f6p	81022008	67544699	1157.9	945.9	746.0	1.22 (1.00–1.54)	0.79
K0_31	f4	290412310	264887288	4781.3	3013.3	2476.9	1.59 (1.51–1.72)	0.82
K0_31	f6p	164736149	146257576	2065.2	1354.0	1197.2	1.52 (1.38–1.67)	0.88
K0_19	f4	17250812	15282173	459.7	358.9	291.3	1.28 (0.97–1.55)	0.81
K0_19	f6p	10226773	8837211	147.2	110.1	89.6	1.34 (0.66–1.89)	0.81
K1_19	f4	30159678	25782851	827.6	650.7	580.1	1.27 (1.00–1.54)	0.89
K1_19	f6p	19117122	16026687	282.9	264.6	214.1	1.07 (0.85–1.72)	0.81
K1_17d7	f4	22448248	19473310	649.8	611.2	562.6	1.06 (0.83–1.27)	0.92
K1_17d7	f6p	10867029	8987079	167.9	170.4	148.3	0.99 (0.81–1.11)	0.87
K1_29	f4	185919270	165390163	3235.8	2039.6	1753.8	1.59 (1.44–1.71)	0.86
K1_29	f6p	112675706	98192767	1497.1	1007.0	846.5	1.49 (1.35–1.77)	0.84
K0_37	f4	652414757	604262254	8779.5	5159.3	4033.1	1.70 (1.56–1.86)	0.78
K0_37	f6p	376487242	342788148	3921.7	2527.9	2160.3	1.55 (1.43–1.76)	0.85

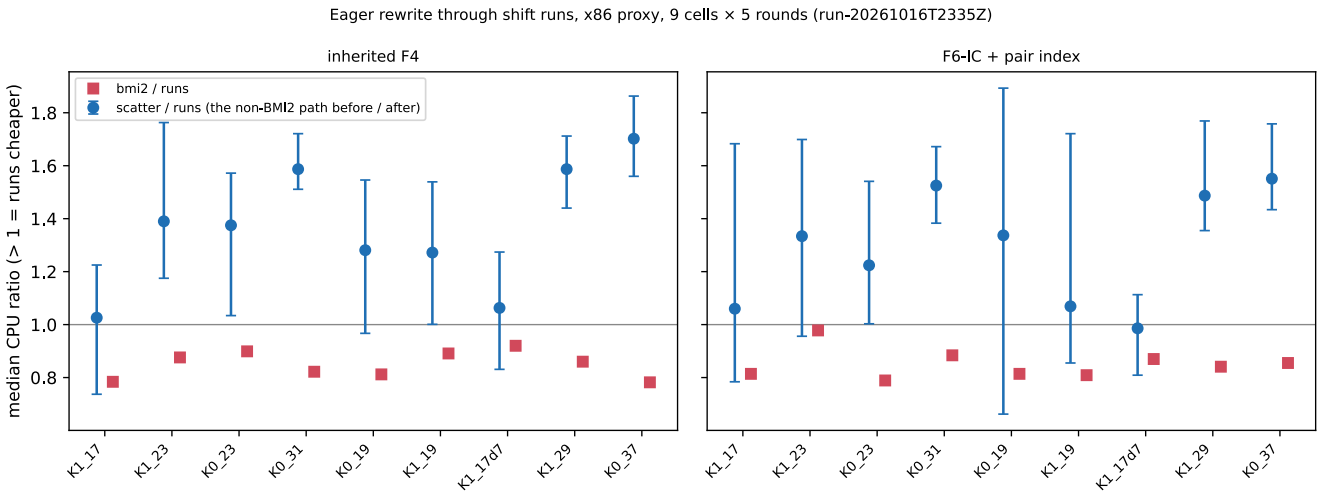


Figure 1: CPU ratios per cell, run-20261016T2335Z, 5 rounds.

Observations

- Against the bit scatter (the previous non-BMI2 path), shift runs: median CPU 1.22–1.70× lower on the six larger cells (K1/2^23, K0/2^23, K0/2^31, K0/2^19, K1/2^29, K0/2^37) on both engines, with the worst round pairing ≥ 1.00 on K0/2^31, K1/2^29, K0/2^37 (both engines) and K1/2^23 f4, K0/2^23 (both); on K1/2^19 1.27× (f4) and 1.07× (f6p); level (0.99–1.06×, brackets spanning 1) on K1/2^17 d6 and d7.
- Against `pdep ∘ pext` on this host, shift runs are 2–28% slower (bmi2/runs 0.78–0.98); the BMI2 path stays the default where BMI2 exists.
- Not measured: an aarch64 host. The shift-run kernel is plain integer code, so the ratio there may differ.

Records

`raw/r<round>-<cell>-<arm>.jsonl`, `raw/*.status`, `host.txt`, `summary.txt`, `table.tsv` (`../../summarize.py`), `fig_runs.svg|png` (`../../plot.py`).